SKYWORTH PV
Residential Solar Solution

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SUN Mate
Introduction

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SUN Ward
Introduction

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SUN Mate Introduction

Adjustable | Aesthetic | Versatile | Cost-effective

Product Overview

Quick to install, Hassle-Free Choice

SUN Mate

Power Range: 900W/1800W

All-in-one Quick-install Solution

Build your power station as easily as assembling building blocks.

A small-scale residential PV solution, perfectly tailored for balcony installations with system capacity under 2kW.

Product Four Features

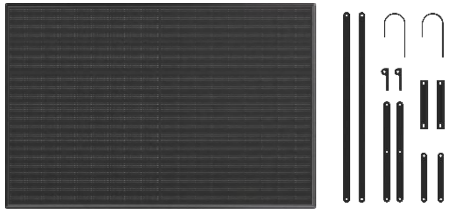
Adjustable

The optimal installation angle can be easily adjusted to suit a variety of installation environments. The power generation efficiency of solar panels can be maximized based on sunlight availability.

Aesthetic

Bauhaus-inspired minimalist design for a sleek, elegant look that seamlessly complements modern home styles.

Flexible



Generate electricity from sunlight anytime and connect to household appliances instantly. Suitable for a wide range of scenarios, the system is compact and easy to store when not in use, with no bulky footprint.

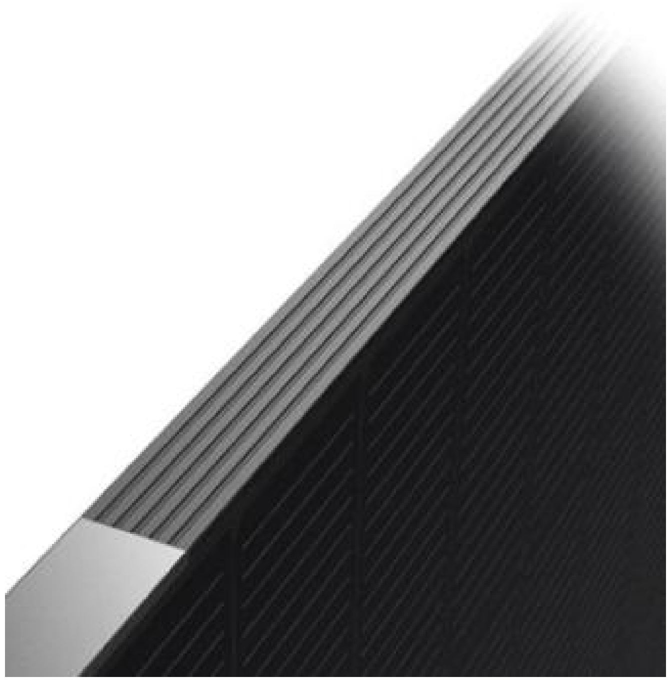
Cost-effective

The standard version is ready to use immediately upon generation and enables free electricity usage during the midday peak period. The energy storage version enables peak-load shifting, perfectly alleviating electricity anxiety during the evening peak period.

Feature: Aesthetic

Bauhaus-inspired Minimalist Design

Meticulously crafted by our Skyworth design team – a Red Dot Design Award winner – to seamlessly integrate with a wide range of modern home styles.



Sunshade Mounting

All-black design (without grid lines)

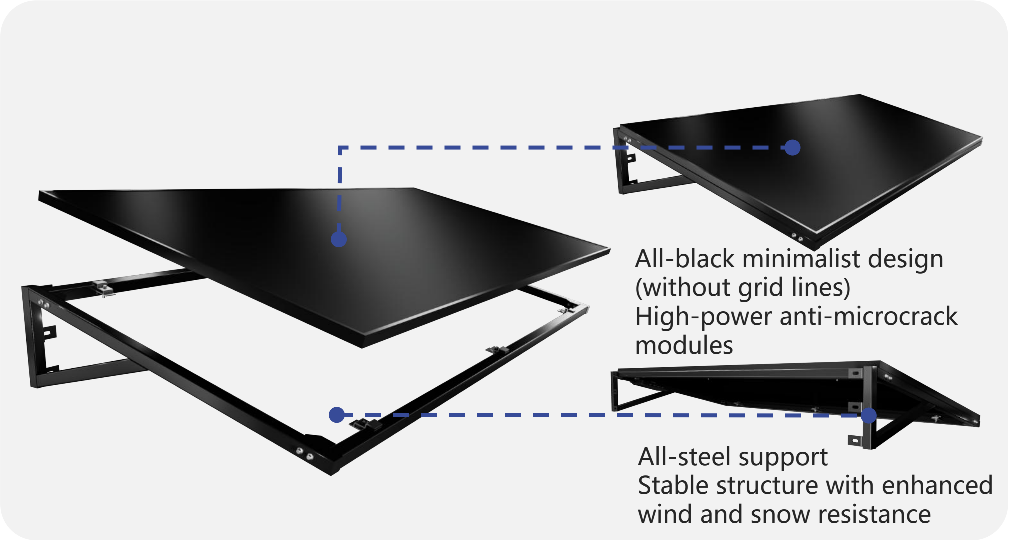
High-power & anti-microcrack

High resistance to wind and snow

Plug-and-play

Sunshade Mounting

Design



Modules and support

Scenes

(Suitable for shop fronts, windowsill exterior facades, and self-built house entrances)

Store fronts

Windowsill exterior facades

Self-built house entrances

Railing Mounting

String-style frame

Rounded corners for collision prevention

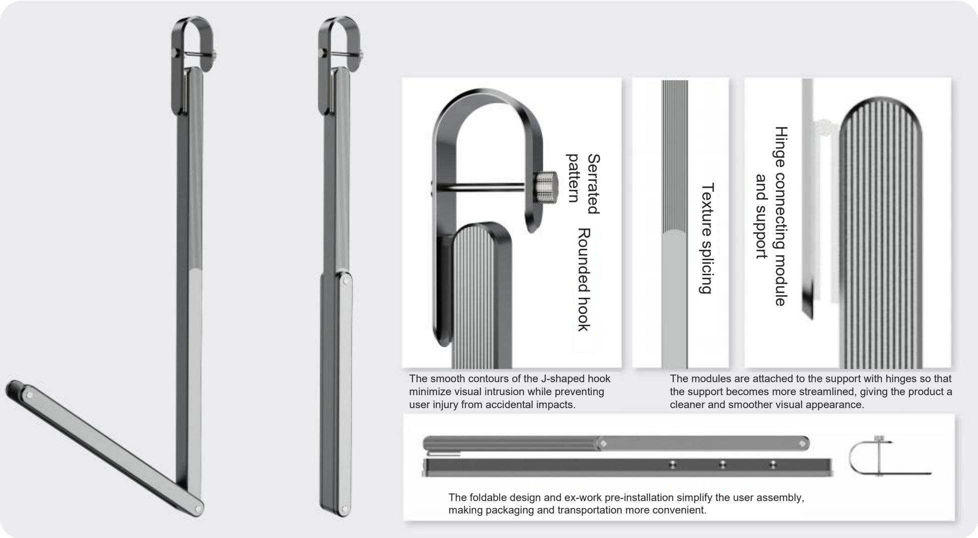
Foldable high-strength support

High resistance to wind and snow

Plug-and-play

Railing Mounting

Design

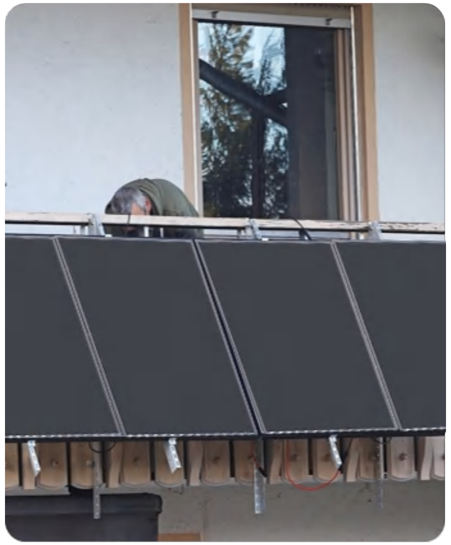


Support: Bauhaus design and foldable support facilitates installation and storage.

Scenes

(Suitable for villas, apartments, and other residences with open balconies)

Villa balcony



Apartment balcony

Utility balcony

Ground Mounting

Minimalist design

Foldable and easy to store

Tool-free three-step installation

Multi-angle adjustable

Plug-and-play

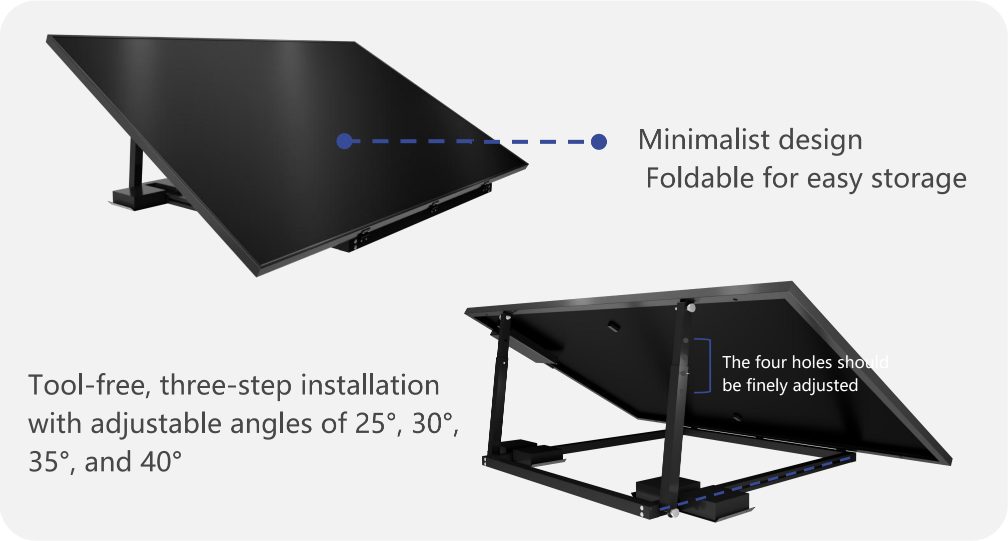
Ground Mounting

Adjustable: 25°, 30°, 35°, 40° Four-angle Adjustment Adapted to optimal sunlight angle in various conditions

The product offers four adjustable angles at 25°, 30°, 35°, and 40° to suit the optimal sunlight angles in different regions and seasons, maximizing solar energy utilization and improving power generation efficiency. It is adapted to different scenarios, orientations, and structures, enabling users to make corresponding adjustments according to the space and balancing practicality and flexibility.

Ground Mounting

Floor mounting design



PV & Mounting

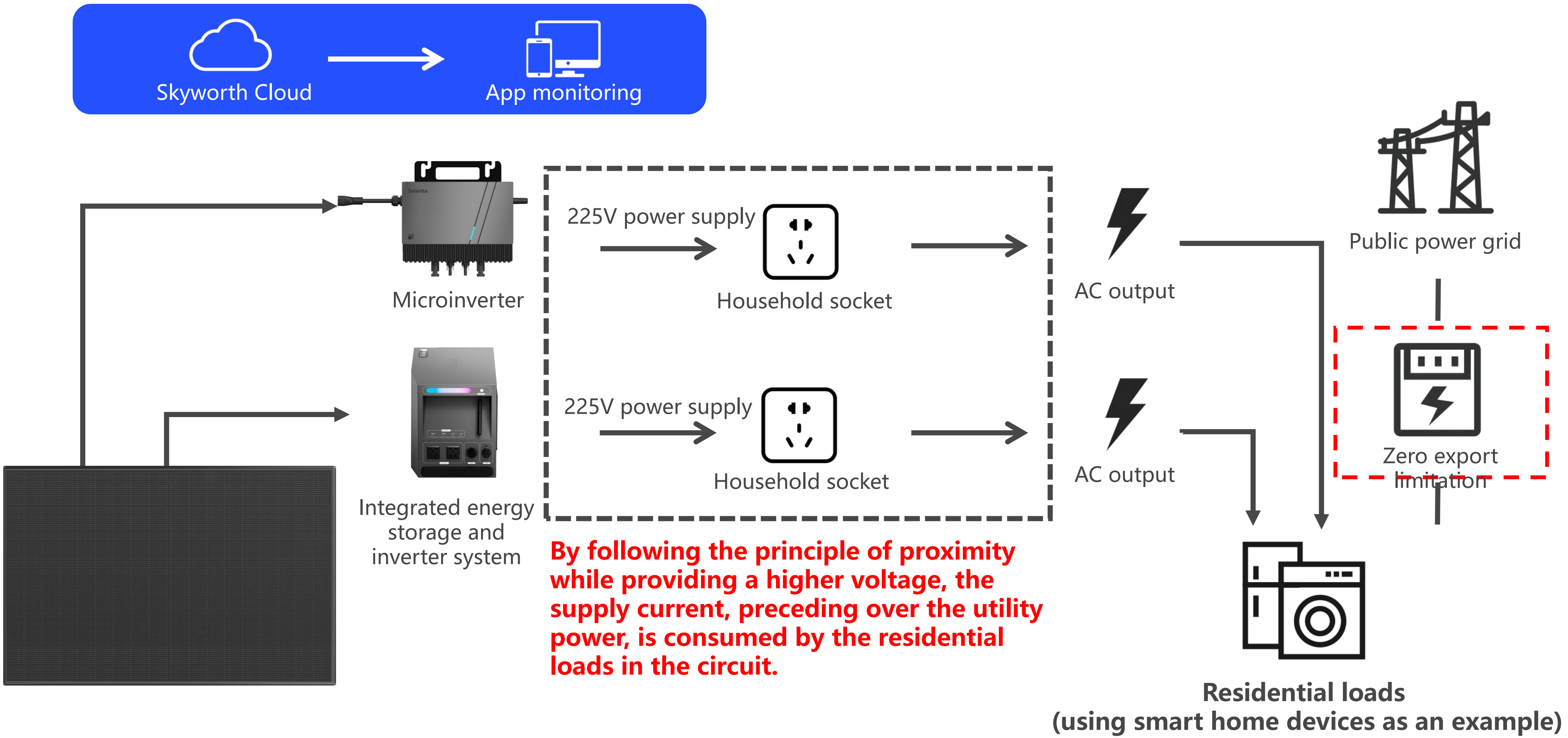
Scenes

Villa courtyard

Self-built house courtyard

camping scene

Feature: Flexible



By following the principle of proximity while providing a higher voltage, the supply current, preceding over the utility power, is consumed by the residential loads in the circuit.

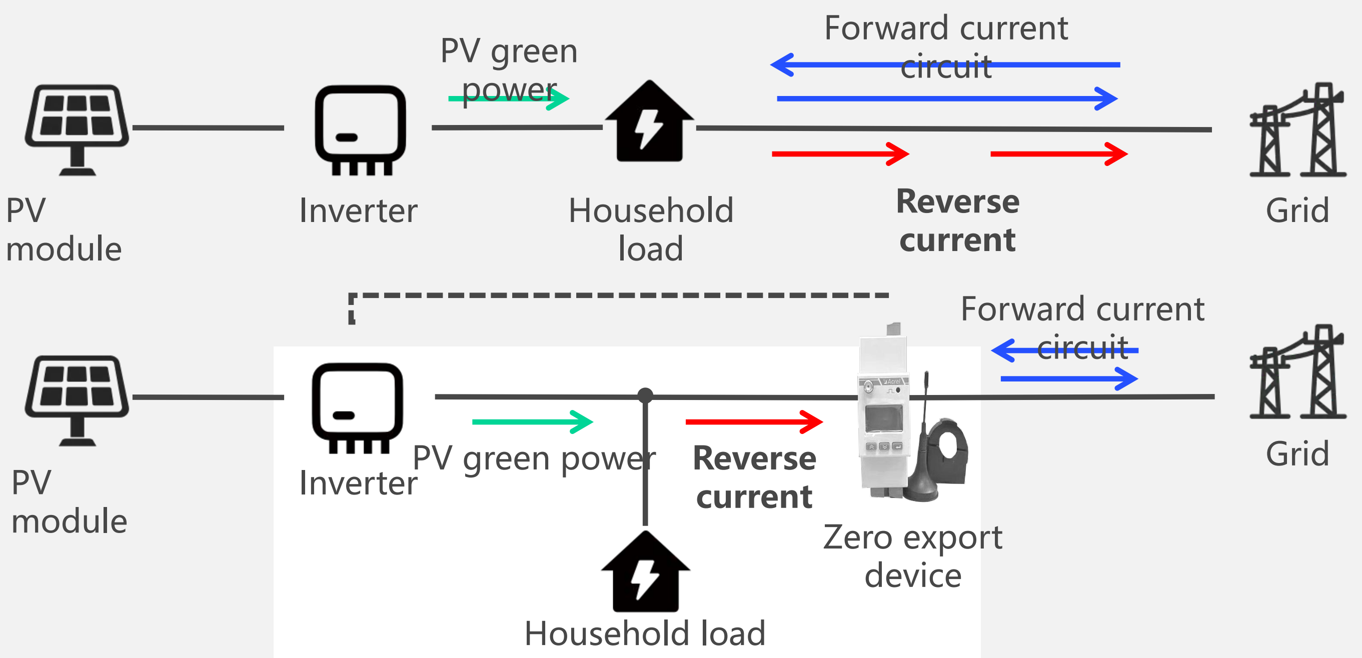
Feature: Flexible

Why Is a Zero Export Limitation Necessary?

Ensure grid safety   Ensure compliant electricity billing

Exclusive supplier providing a zero export device
Adaptive adjustment for stable returns
Fully eliminate the risk of reverse power flow
Integrated AI-powered monitoring technology

Zero-export Management Principle



The zero export limiter continuously monitors whether current is being exported and dynamically adjusts the inverter’s output power to match real-time load consumption

Zero-export Management Solution

Precise zero-export management

Second-level rapid current response

Completely eliminate power grid fluctuations caused by export

Adaptive adjustment

Built-in dynamic power regulation module

Adjust the zero-export management threshold according to actual electricity usage

Integrated monitoring

Integrated local network/Wi-Fi communication module

Feature: Flexible
Smart Operation

Control at your fingertips, anytime, anywhere

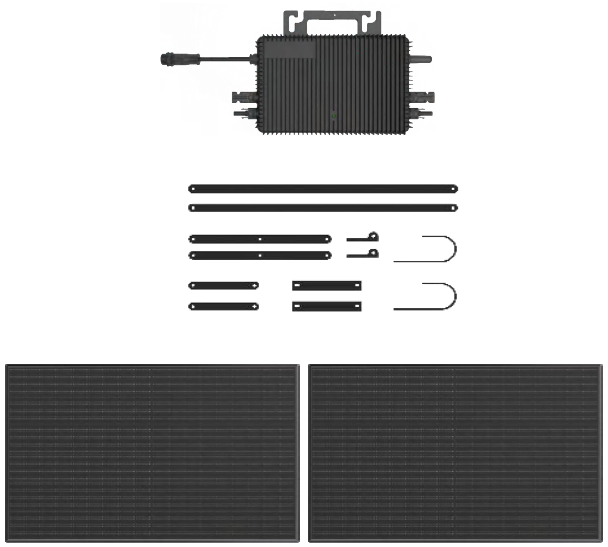
- Full visibility into your rooftop PV station status
- One-tap check of daily/monthly/annual generation and revenue
- Dual-view upgrade: mobile & desktop full access
- AI O&M for worry-free management

Modular Design

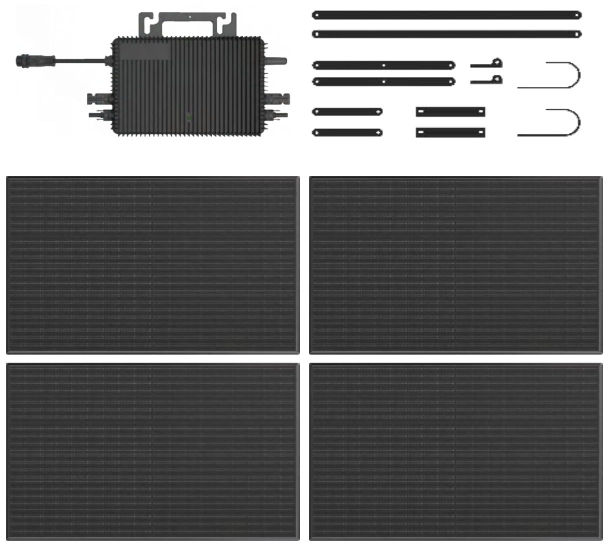
A self-installation kit
Three major components
- **Modules**
- **Microinverter**
- **zero export device.**

Notes: Installation in Thailand requires hardwired connection to the AC distribution box—no plug-and-play allowed

1-to-2



1-to-4



Type	Series	Series Model	System Capacity(Wp)	Microinverter(W)	Export Limitation Device
RAILING MOUNTING	1-to-2	SWM800-R	900	800	Standard
	1-to-4	SWM1600-R	1800	2000	Standard
SUNSHADE MOUNTING	1-to-2	SWM800-S	900	800	Standard
	1-to-4	SWM1600-S	1800	2000	Standard
GROUND MOUNTING	1-to-2	SWM800-G	900	800	Standard
	1-to-4	SWM1600-G	1800	2000	Standard

Feature:Cost-effective

1-to-2 (900W)

Daily Power Generation
3.0kWh

TV	*2	15H
Light Bulbs	*6	8H
Refrigerator	*1	20H

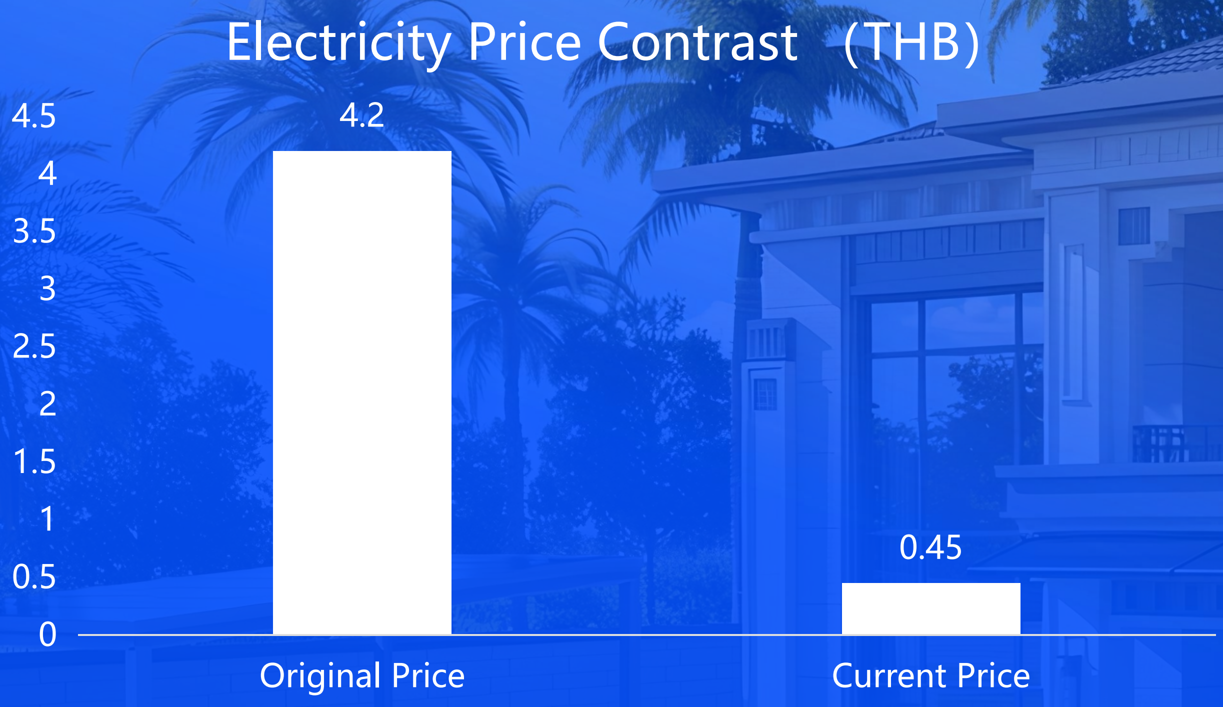
1-to-4 (1800W)

Daily Power Generation
6.0kWh

Light Bulbs	*6	16H
Refrigerator	*2	20H
AC	*1	7H

1P AC 600-800W

Feature: Cost-effective



Electricity cost as low as

0.45 THB/unit

One-time investment
35,999 THB

Total 30-year return
331,128 THB

9.2x
Return on Investment

Feature Four: Cost-effective

Example: Monthly electricity consumption of 288 kWh Unit
(electricity rate: 4.23 THB/kWh)

1-to-2 900W

2 panels

Generates 3.6 kWh of energy via SUN MATE

Energy savings = Power × hours × 30 days × rate

Daily power can run an air conditioner for 5.3 hours

Monthly savings

459

THB on electricity

Example: Monthly electricity consumption of 720 kWh
(electricity rate: 4.23 THB/kWh)

1-to-4 1800W

4 panels

Generates 7.2 kWh of energy via SUN MATE

Energy savings = Power × hours × 30 days × rate

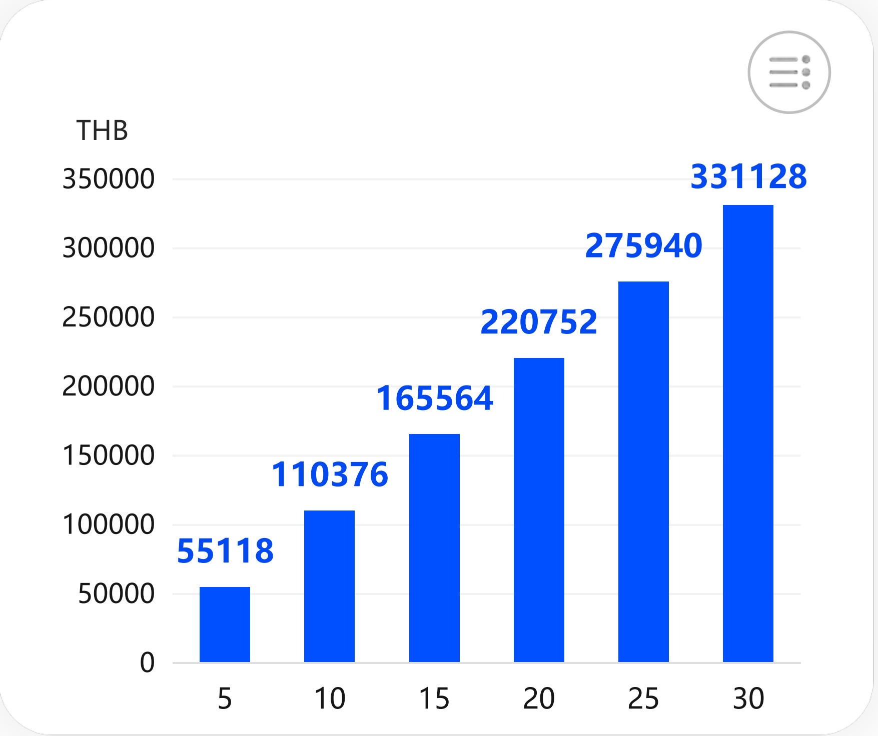
Daily power can run an air conditioner for 5.7 hours

Monthly savings

919

THB on electricity

1-to-4 1800W



ROI Calculation

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SUN Ward Introduction

SUN Ward Overview

Unique & Tailor-Made

SUN WARD

Power Range 2kW-30kW

Customized Photovoltaic Solution

Tailored for medium to large residential PV scenarios

Integrated design, construction, and O&M

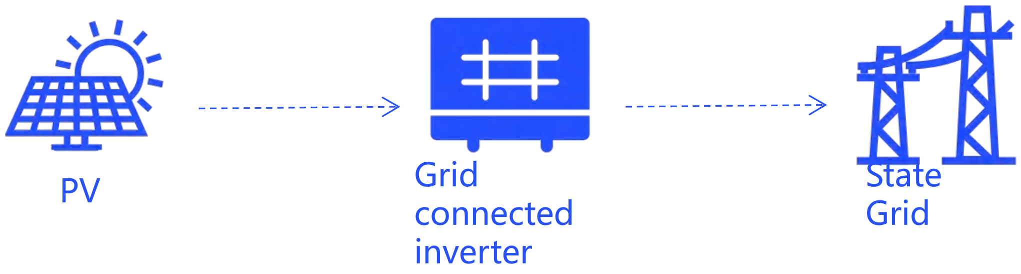
Compatible with both flat and pitched roof configurations

SUN Ward Scenario

Scenario 1: On-Grid PV System

This is the most basic application scenario of PV power generation systems.

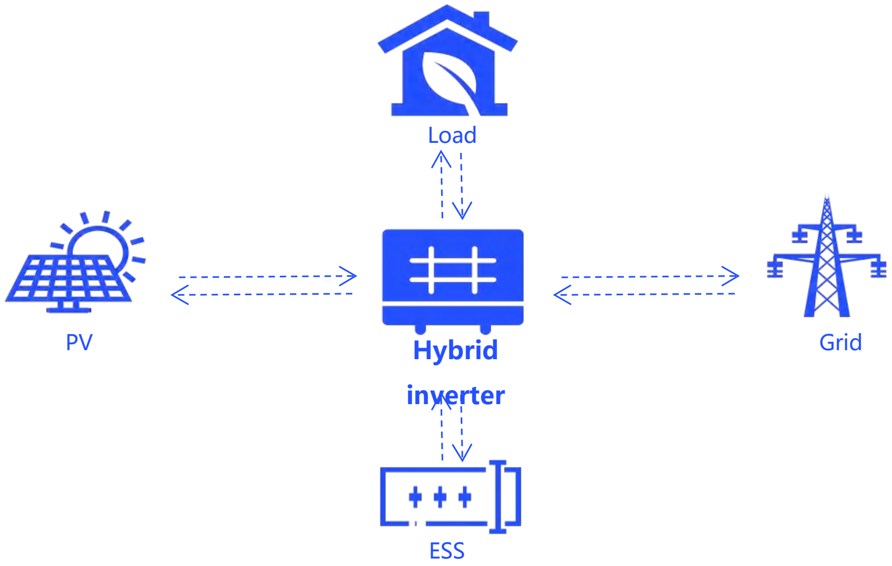
Grid-tied PV systems without energy storage are the global mainstream for residential installations. They feature an asset-light model with high adaptability, making them ideal for areas with a stable, mature grid and no requirement for backup power.



Scenario 2: PV&ESS PV System

This is a power station solution scenario that coexists with mains power.

On-grid PV plants with added energy storage form composite power facilities, featuring grid connection plus energy storage regulation. The PV-storage synergy retains photovoltaic’ s clean energy attributes and offsets its unstable output, serving as a key development model for high penetration of new energy into grids.



SUN Ward Scenario

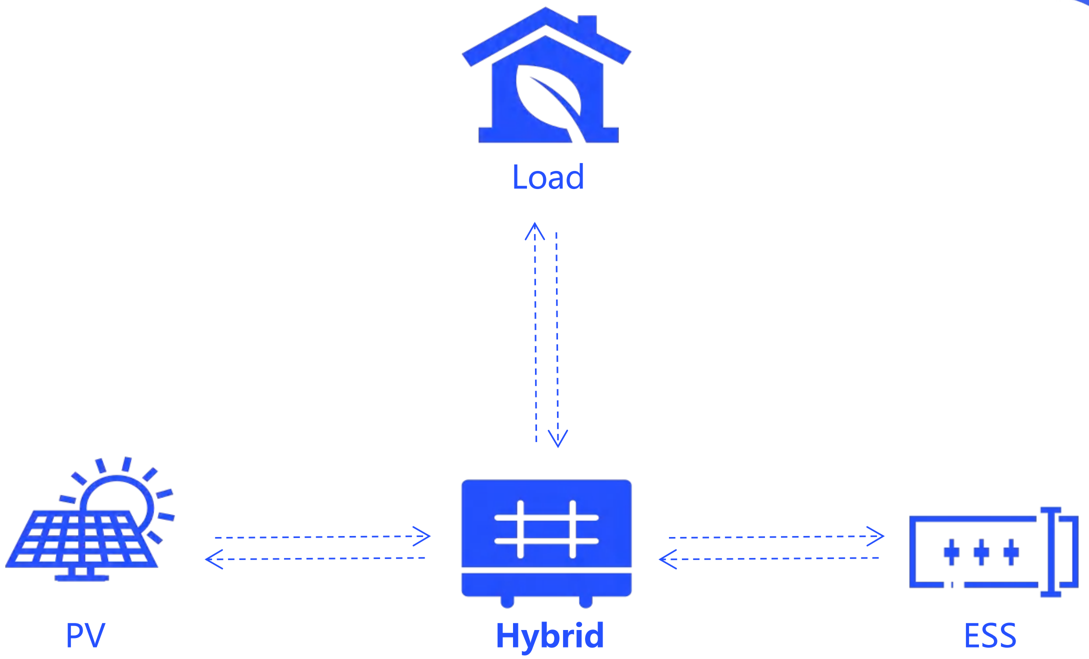
Scenario 3: Off-grid PV system

Independent power supply scenarios that are disconnected from the power grid

1. Direct PV Power Supply Mode
Sufficient daylight: PV power directly powers the load, surplus charges the energy storage battery.

2. Battery Power Mode
Night/insufficient PV power (cloudy/rainy): battery alone or with PV system jointly powers the load.

3. Hybrid Power Supply Mode
System paired with grid/diesel generator: when battery is low, grid/diesel generator powers the load.



SUN Ward Roof Style:
Flat and Sloped Roof Scenarios

SUN Loft Core Advantages

Multiple Functions

Rainproof, moisture-proof, cooling, and heat-insulating.

Flexible Space

2.7m high support creates ample leisure space for a sky garden or recreational area.

Reliable & Durable

TÜV Rheinland certified: corrosion-resistant, moisture-proof, and typhoon-resistant (Grade 12).

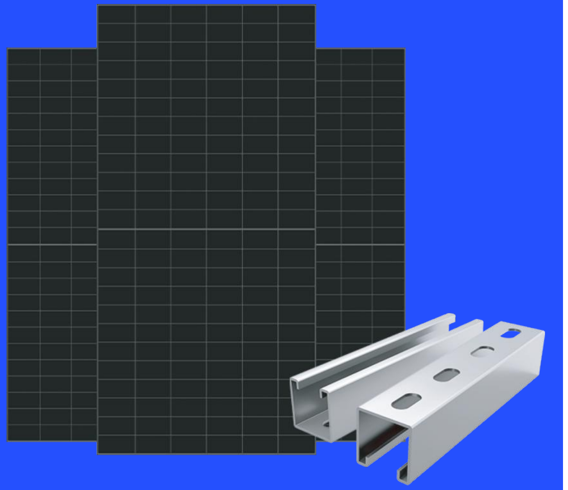
Maximized Revenue

5° gable design ensures year-round sunlight, boosting power generation and revenue.

One-stop Solution

High Energy Yield

- High Efficency N-type PV Panel
- Stable Output at High Temperatures



PV&Mounting

High Efficiency

- Max. Efficiency 98.6%
- 1.5X DC Overmatching
- Dual MPPT Design,
- Fault Redundancy
- Stability Enhancement

Inverter

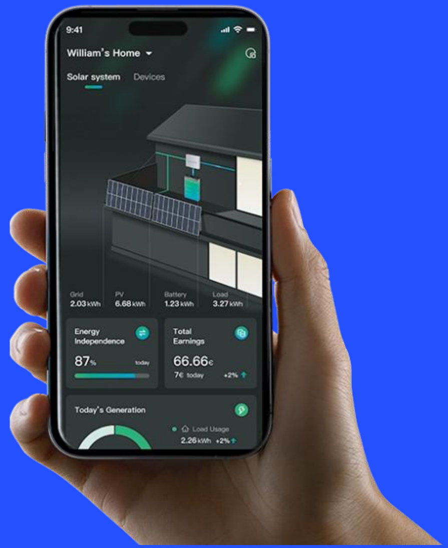
Reliable Battery

- Premium Cells, $LiFePO_4$ batteries with multi-layer safety protection
- 10-Year Warranty

Battery

Smart Operation

- One-Stop Control: Manage PV, storage, heat pumps & EV chargers in one.
- Smart Savings: Optimize tariffs & lower electricity costs.



Solavita Cloud APP

High-Quality

Exceeding Industry Standards for a 30-Year Lifespan

Energy Bright Technology builds PV plants to above-industry standards. Rigorous internal control and professional external supervision ensure standardized engineering services. This dual guarantee extends the station’ s lifespan, enabling stable, efficient power generation for up to 30 years.

High-Standard Quality Management

Material Selection → Survey & Design → Project Management → Station Acceptance → O&M Services

Material Selection
- 318 above-industry standards
- TÜV Rheinland supervision

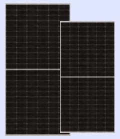
Survey & Design
- UAV 3D modeling
- Roof load assessment

Project Management
- Visualized construction
- Standardized processes

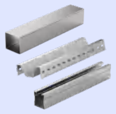
Station Acceptance
- Owner acceptance
- Grid acceptance
- O&M acceptance

O&M Services
- Smart O&M
- 7×24 all-weather service

Selected Premium Products



Modules
30-year power output warranty
Product Warranty: 15 years



Mounting
30-year anti-corrosion warranty

Inverter
25-year design life
Product Warranty: 10 years

Value-Added Revenue

Smart O&M: Securing 30 Years of Steady Revenue

Energy Bright Technology operates an all-weather smart O&M platform, delivering 2-hour rapid response and 24/7 monitoring to provide users with end-to-end O&M management services.

- Plant Monitoring
- Fault Resolution
- Equipment Maintenance
- Inspection Management
- Cleaning Services
- Insurance Services

*This service is optional for full-payment customers.

SUN Ward Return Calculation

Make Every Ray of Sunshine Count

Option 1

8,278 kWh PV power generation/Year

Item	Value
System Solution	8,278 kWh Annual Generation
Self-Consumption Ratio	80%
30-Year Power (kWh)	230961
Initial Investment (THB)	114999
Payback Period (Years)	4.4
LCOE (THB/kWh)	0.62
30-Year Profit (THB)	776031

Option 2

8,278 kWh Annual PV Generation/Year +5kWh Energy Storage

Item	Value
System Solution	8,200 kWh Annual Generation+5kWh ESS
Self-Consumption Ratio	100%
30-Year Power (kWh)	230961
Initial Investment （THB）	135000
Payback Period (Years)	4.1
LCOE (THB/kWh)	0.58
30-Year Profit (THB)	970039

ROI Calculation

SUN Ward Overview

Unique & Tailor-Made

SUN WARD PRO

Power Range **30kW+**

Customized Photovoltaic Solution

Tailored for medium to large residential PV scenarios

Integrated design, construction, and O&M

Compatible with both flat and pitched roof configurations

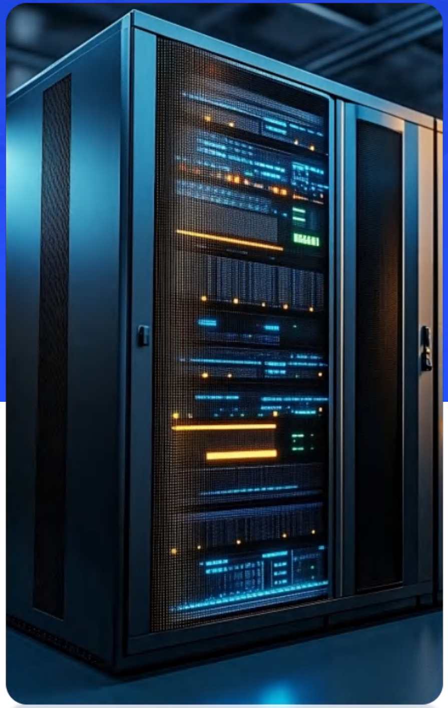
SUN Ward Pro Overview

- Power range covering 30 kW and above, up to megawatt scale
- Suitable for manufacturing, warehousing & logistics, and commercial buildings
- Predictable power generation, more controllable returns
- Prioritize self-consumption to reduce reliance on high-cost grid electricity

Medical Buildings

Commercial Buildings

Digital Infrastructure



Industrial Parks

Solar PV Charging Carports

SUN Ward Pro Return Calculation

Taking a 150kW project in Bangkok, funded entirely by the property owner, as an example, it uses an original system with SKYWORTH 630W modules.

Installed capacity	Annual effective sunshine	Self-consumption ratio	Average electricity tariff
150kW	1460小时	80%	4.2THB

Project investment

System unit price	20.25 THB/W	initial investment	3,037,500 THB

Power generation calculation

Average annual power generation	175,200 kWh	25 years of power generation	4,073,400 kWh

Return calculation

Annual power generation income	684,000 THB	25-year power generation income	17,110,000 THB	Cost of electricity	0.62 THB

ROI

Pay Back	4.4 years

*The actual returns of a project are related to multiple factors such as local climate, sunlight conditions, roof conditions, enterprise electricity consumption curves, and electricity prices. The above returns calculations are for reference only.

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